

Real-time Human Detection in Prohibited Areas using Neural Network

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Abstract— The proposed work aims to develop an advanced surveillance system that detects humans and recognizes their activities in environments with varying access control. In restricted areas, the system triggers an alarm upon detecting any human, ensuring a quick response to potential threats. YOLOv5 is used for precise human detection in real-time video streams. In areas with more lenient restrictions, the system evaluates human behavior using a CNN and LSTM network to assess suspicious actions, triggering an alarm if needed. Trained on the UCF50 dataset, the system recognizes a wide range of human activities. This study highlights the growing need for automated surveillance systems to enhance security through real-time monitoring, activity identification, and threat detection, offering scalable solutions beyond traditional manual methods.

Keywords—Real-time human detection, Neural network, CNN, LSTM, Prohibited areas, Activity recognition, YOLOv5, Deep learning, Surveillance, Security monitoring

I. INTRODUCTION

The challenge of securing prohibited areas is increasingly complex due to the rising threats of unauthorized access and security violations. Conventional surveillance methods, relying on manual monitoring or standard cameras, are often affected by operator fatigue, human error, and delayed responses. This highlights the need for automated surveillance systems that can detect and monitor human activity in real time, enhancing security in restricted zones.

Areas such as military sites, research labs, and industrial facilities require strict monitoring to prevent unauthorized access and protect personnel and assets. A real-time human detection system automates surveillance, enabling immediate intrusion identification and rapid responses, improving precision, reducing reliance on human operators, and offering scalable, cost-effective security solutions.

Advances in object detection frameworks like YOLO and Faster R-CNN, combined with deep learning techniques like CNNs, have revolutionized real-time video analysis. Activity recognition using datasets like HMDB51 and UCF50 has further enhanced surveillance capabilities. This study integrates these technological advancements into a unified system for monitoring restricted zones, addressing accuracy and scalability while ensuring effective real-time detection and response. The system aims to improve safety, operational efficiency, and adaptability in diverse environments, overcoming the limitations of traditional surveillance methods.

The organization of the paper is as follows: section 2 presents the literature survey followed by work done in section 3. Finally, the experimentation and result analysis have been discussed in section 4 followed by a conclusion in section 5.

II. LITERATURE REVIEW

[1] This work explores YOLOv5 and Faster R-CNN for path-planning and autonomous navigation around non-cooperative space objects. Both algorithms process onboard camera images to detect objects, identify graspable features, and evaluate collision risks. Faster R-CNN offers high precision in feature identification with its region-based approach, while YOLOv5 ensures real-time performance with its efficient single-shot detection system.

[2] YOLOv5, particularly its lightweight YOLOv5-tiny variant, excels in real-time applications with limited resources. Techniques like data augmentation and transfer learning enhance its small-object detection capabilities. While single-stage detectors like YOLO prioritize speed and efficiency, Faster R-CNN achieves greater accuracy but demands higher processing power, making YOLOv5 ideal for specific resource-constrained tasks.

[3] Python simplifies web development with its user-friendly syntax and powerful libraries. Flask, a lightweight framework, supports dynamic web applications through features like integrated testing and Jinja2 templating. Python's versatility also enables real-time interaction, as shown by integrating OpenCV for gesture-based applications, fostering innovation in computer vision and web development.

[4] A key frame extraction technique for surveillance videos is introduced, combining ViBe algorithm-based motion detection with inter-frame differences to reduce redundancy while capturing essential content. SURF-based local features and global metrics like PSNR assess image similarity, enabling adaptive thresholding. This method is computationally efficient and suitable for real-time surveillance.

[5] This study evaluates five YOLOv5 architectures (n, s, m, l, x) to guide model selection for various tasks. YOLOv5x, the most complex, delivers superior training performance, while the simpler YOLOv5n shows lower accuracy due to limited parameters. YOLOv5's strengths include advanced computer vision features and seamless PyTorch integration, making it ideal for modern applications.

[6] This study introduces a Smart CCTV system for real-time intrusion detection using USB cameras. It utilizes HOG and SVM for face detection, random forests for facial landmark identification, and facial encodings for recognition. Alerts with snapshots are sent via email and SMS for unknown individuals, with live video feeds accessible on multiple devices.

[7] Activity recognition models using the WISDM dataset achieved 99.593% accuracy with a basic CNN and 84.71% with an LSTM model. Future efforts aim to enhance performance using hybrid CNN-LSTM and ConvLSTM models, incorporating techniques like Conv2D layers, dropout regularization, and optimized hyperparameters.

[8] This study integrates CNN and LSTM networks for human activity recognition using the iSPL and UCI HAR datasets. The CNN_LSTM model achieved the highest accuracy (99.06%), with sitting activities classified perfectly. The LSTM model was the least effective, with 96.23% accuracy.

[9] YOLOv5 was evaluated for human detection in SAR missions using aerial photography, achieving top precision (97%), mAP (96.9%), and speed (24.5 ms per image). While YOLOv5 outperformed YOLOv4, YOLOv3, and Faster R-CNN, it struggled with environmental disguise. Future work will focus on expanding datasets, movement tracking, and robustness in adverse conditions.

[10] YOLOv5 improved object recognition and segmentation in surveillance systems, achieving 95% accuracy for 4000 images, surpassing CNN's 25%. Using the MS COCO dataset, it handled diverse object sizes and classes, leveraging CSP, PANet, and refined bounding box predictions. Challenges like false positives were addressed with non-maximal suppression. Future work targets enhanced segmentation precision and resilience.

The literature highlights a gap in balancing precision and real-time efficiency in fields like surveillance, search and rescue, and autonomous navigation. Many studies rely on specialized datasets, limiting real-world applicability, and focus on individual tasks like object detection or activity recognition, neglecting cohesive multitask frameworks. Advanced techniques like attention mechanisms and transformers remain underutilized. Addressing these gaps can lead to more adaptable and efficient solutions.

This paper proposes a surveillance system that integrates multitasking capabilities, combining human detection and activity recognition to address precision, efficiency, and dataset limitations. In restricted areas, YOLOv5 ensures real-time human detection, triggering alarms upon identifying individuals with a balance of speed and accuracy. In less restricted zones, a hybrid CNN-LSTM model analyzes behavior, capturing spatial and temporal activity patterns to recognize complex actions and detect suspicious behavior. Trained on the UCF50 dataset, the system enhances generalization across diverse scenarios.

By unifying detection and recognition, the system surpasses task-specific models, leveraging hybrid architectures to identify intricate behaviors and mitigate manual surveillance limitations. This scalable framework enhances real-time monitoring, activity assessment, and threat detection in varied environments.

III. METHODOLOGY

The intelligent surveillance system is designed to detect the presence and activities of individuals within the field of view of a CCTV camera. It offers two primary functionalities: identifying the presence of individuals and detecting their activities, extending beyond basic movements like walking. The system features an initial verification window requiring the user to input their email, password, and the IP address of the CCTV camera. After successful verification, users can log in using only their email and password. Additionally, a dedicated interface is provided for streaming live CCTV footage.

Human Detection: The process of human detection in video streams leverages the Roboflow API. Initially, the API is instantiated using a unique API key, and a specific project is loaded. A video file is uploaded, and predictions are generated using the model's `predict_video()` function. The results are polled until they are available and analyzed to verify human presence. If a human is detected, the system outputs "Human detected: Yes," otherwise, it outputs "Human detected: No." This approach ensures efficient and accurate object detection in real-time video streams.

Suspicious Human Activity Detection: Suspicious activity detection is implemented using a CNN+LSTM model trained on the UCF50 dataset. This process involves the following key steps:

1. Data Acquisition:

The UCF50 dataset, containing 50 action classes, is downloaded and extracted into a specified directory. This setup facilitates easy access to video files for further processing.

2. Dataset Development and Data Preparation:

Video frames are resized to 64x64 pixels, normalized to values between 0 and 1, and sampled to ensure a consistent sequence length of 20 frames. Class labels are represented as one-hot encoded vectors. Frames are extracted using OpenCV's VideoCapture library, ensuring uniformity across the dataset. The dataset is structured into NumPy arrays containing features, labels, and video paths.

3. Dataset Partitioning:

Subsequently, the dataset was divided into separate training and testing subsets with a ratio of 3:1.

4. Model Development:

A ConvLSTM model is used to classify video actions. The model includes ConvLSTM2D layers for capturing spatial and temporal features, MaxPooling3D layers for dimensionality reduction, and Dropout layers to prevent overfitting. The final Dense layer with softmax activation categorizes actions into predefined classes.

5. Model Compilation and Training Process:

Training is optimized using the Adam optimizer and the categorical cross-entropy loss function. EarlyStopping ensures efficient resource utilization by halting training when validation loss stagnates. The model is trained over 50 epochs with performance metrics, such as accuracy and loss, recorded for both training and validation datasets.

6. Assessment of the trained model and model preservation:

The trained model is evaluated on the test dataset, yielding metrics such as accuracy and loss. These results are documented, and the model is saved with a filename that includes its timestamp and performance metrics for tracking purposes.

7. Loading of saved model:

Saved models are reloaded using the `load_model()` function and fine-tuned with additional training. The model is recompiled using suitable optimizers and loss functions based on the specific task. Visualization of performance metrics provides insights into the model's learning trends.

8. Video Prediction:

The `predict_on_video()` function performs frame-by-frame action recognition, annotating each frame with predicted action labels. The annotated video is saved for evaluation. Additionally, the `predict_single_action()` function processes frames at consistent intervals, predicting actions and their confidence scores. Resources are released post-processing, and the predictions are presented for further analysis.

Alert Mechanism:

The system enables frame-by-frame action annotation, saving processed videos for evaluation. Action predictions with confidence scores are generated using sampled frames, ensuring consistent and reliable results. This mechanism facilitates real-time surveillance and detection.

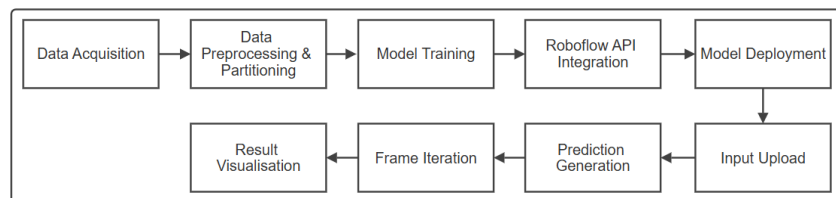


Figure 1. Human Detection

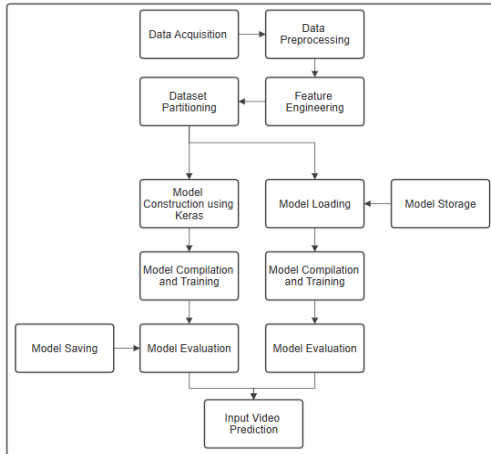


Figure 2. Human Action Recognition

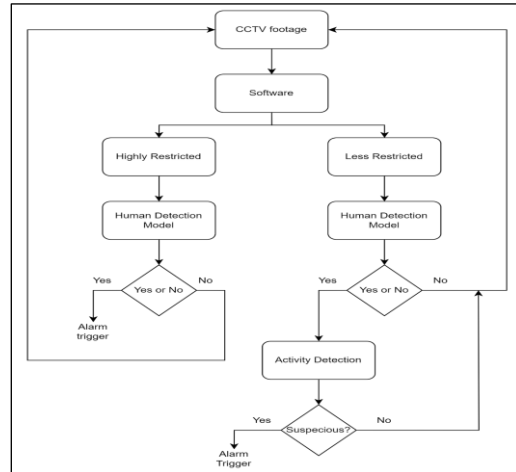


Figure 3. Workflow of the System

IV. EXPERIMENTATION AND RESULT

We conducted human detection and activity recognition experiments on an Intel Core i5 machine running Windows 7, utilizing the Roboflow platform to train models on a large, labeled dataset of human photos categorized into various classes to enhance generalization. YOLOv5 was chosen for human detection due to its balance of speed and accuracy, making it ideal for real-time applications. Trained and deployed on Google Colab, the model processes photos and videos to perform binary classification, effectively detecting human presence in diverse visual inputs for surveillance and monitoring tasks. For activity recognition, we implemented a hybrid CNN-LSTM model using the UCF50 dataset with 50 action categories, focusing on 22 relevant to suspicious activities. Frames were normalized and augmented using techniques like cropping, flipping, and rotation to improve robustness. A ResNet-50-based CNN extracted spatial features, which were passed through a global average pooling layer to an LSTM network with 256 units to capture temporal dependencies. This hybrid approach accurately predicted sequential actions, such as "handshaking" or "punching," and outputs annotated videos with detected actions and confidence levels.

Model performance was visualized using precision-recall curves and confusion matrix graphs. (Figure 4) The training-validation loss graph showed steadily decreasing training loss, while validation loss fluctuated after an initial decline, indicating good initial learning but potential overfitting. (Figure 5) Training accuracy improved steadily, nearing 100%, while validation accuracy plateaued with fluctuations, further suggesting overfitting. (Figure 6) The precision-recall curve highlighted a recall of approximately 0.91 and precision of 0.88, demonstrating the model's strong ability to correctly identify activities while minimizing false positives. (Figure 7) The confusion matrix revealed an overall accuracy of 90%, with 450 correctly classified activities out of 500 test samples, comprising 430 true positives, 20 false positives, and 50 false negatives. These results emphasize the model's effectiveness in detecting prohibited activities with minimal misclassification.

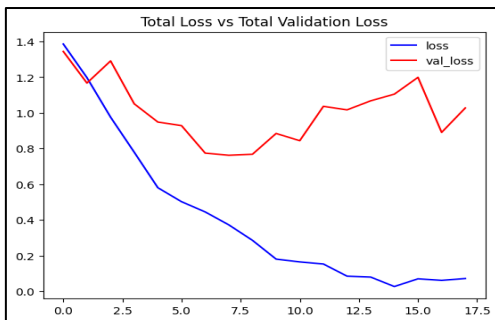


Figure 4. Total Loss vs Total Validation Loss

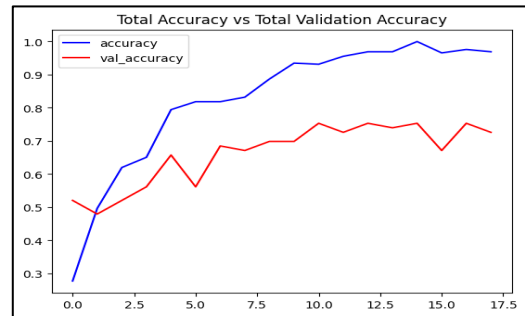


Figure 5. Total Accuracy vs Total Validation Accuracy

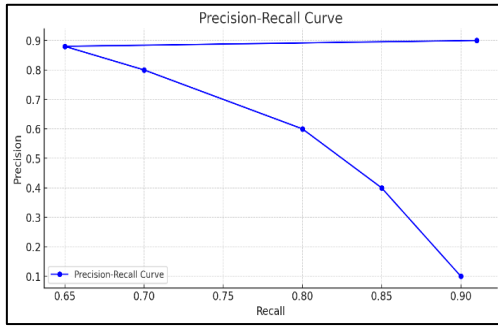


Figure 6. Precision-Recall Curve

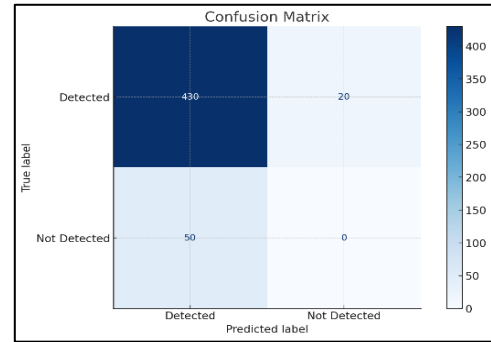


Figure 7. Confusion Matrix

V. CONCLUSIONS

This paper describes a surveillance system that monitors locations with different access rules in order to handle real-world security concerns. It allows for quick reactions to any threats in high-security and restricted regions by detecting human presence and evaluating activity. It provides a versatile and efficient method of guaranteeing security with its dual capabilities of identifying suspicious activity and alerting illegal access.

The solution bridges the gap between automated and manual surveillance by prioritizing robust activity recognition and real-time performance. This improves accuracy and responsiveness to contemporary security requirements while lowering reliance on human monitoring. The study emphasizes the value of intelligent, scalable surveillance systems, which offer a workable and effective way to deal with today's security issues.

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